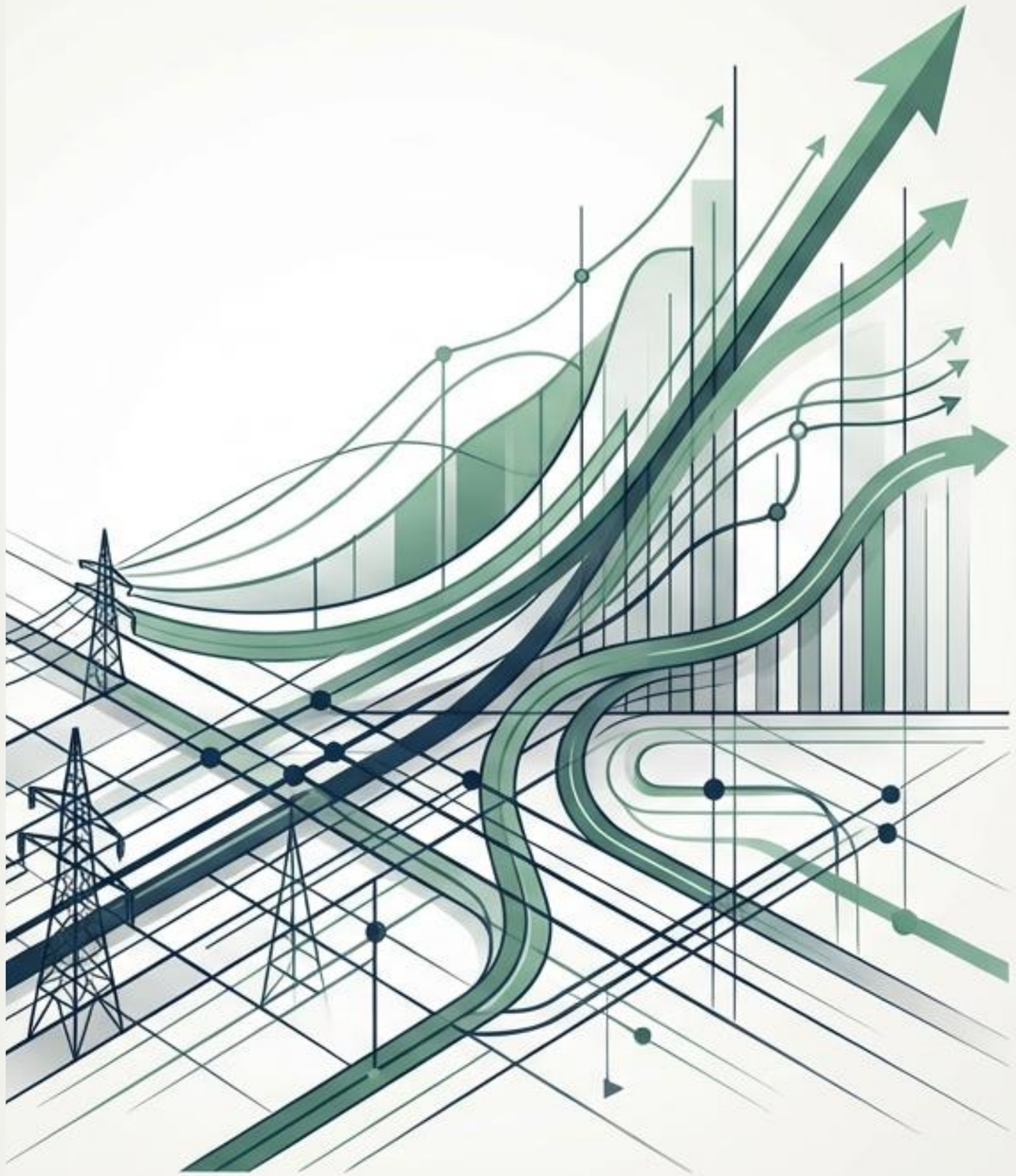
PROJECT GRIDSHIELD

Stochastic financial optimisation of load forecasting.

Securing Lumina Energy's financial and regulatory posture under Availability Based Tariff escalation.



Securing the grid and protecting the bottom line

The Mandate

Minimise financial exposure under Maharashtra's Availability Based Tariff (ABT) using 15-minute resolution, 2-day ahead forecasting.

The Threat

Severe, asymmetric penalties where under-forecasting is penalised up to 3x more heavily than over-forecasting, compounded by post-COVID volatility.

The Engine

A dual-quantile Machine Learning architecture (LightGBM) built on Newsvendor optimisation and strict regime modelling.

The Victory

Reduced baseline financial exposure by 34% in Stage 1, successfully absorbed a ₹6/kWh peak penalty shock, and achieved 100% compliance with Stage 3 Board Directives.

The asymmetric penalty trap

The Traditional Flaw

Standard models (ARIMA, OLS) target average accuracy (0% bias). Under ABT rules, this guarantees financial bleeding.

The Stage 2 Penalty Reality

Actual > Forecast (Under-forecasting): ₹6/kWh (Peak) / ₹4/kWh (Off-Peak)

Forecast > Actual (Over-forecasting): ₹2/kWh

The Strategic Shift

We must stop optimising for statistical error (RMSE) and start optimising for financial risk.

Missing a peak forecast by 1 MW costs three times more than over-procuring by 1 MW.

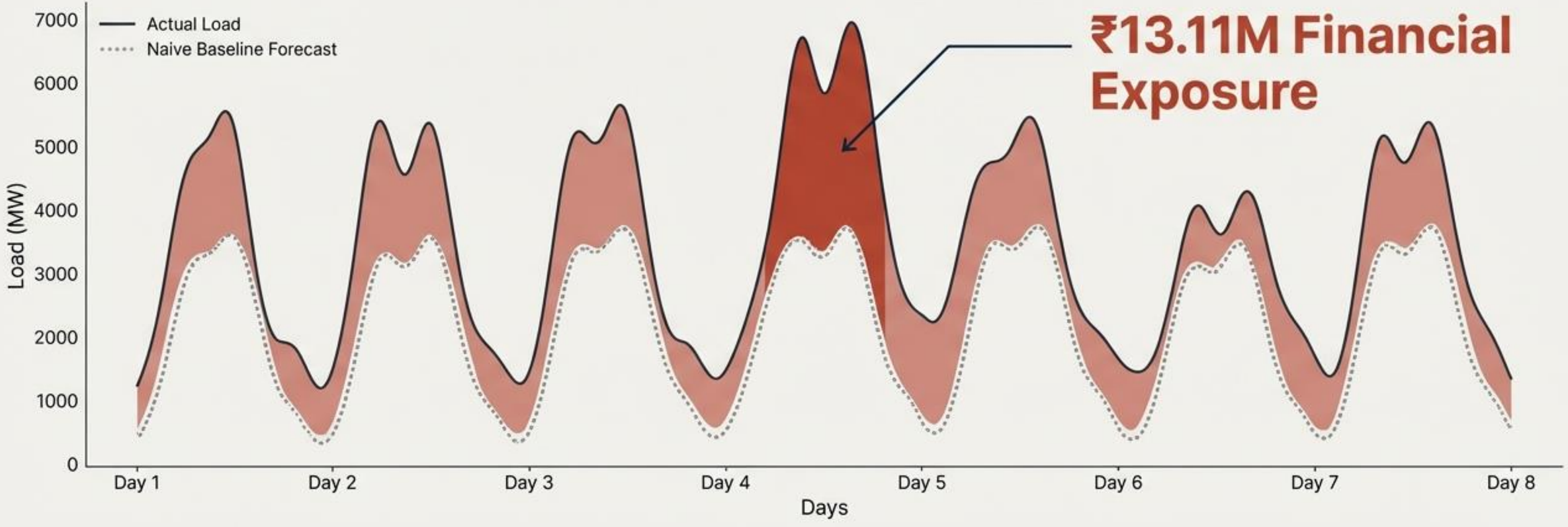
The financial bleed of the status quo

The Naive Approach: Assuming this week's load will match last week's load.

The Cost: Simulating this baseline exposed the grid to ₹13.11 million in deviation penalties.

Root Causes for Failure:

- Total inability to adapt to intra-day weather shocks.
- Failure to capture regime shifts altering electricity elasticity.

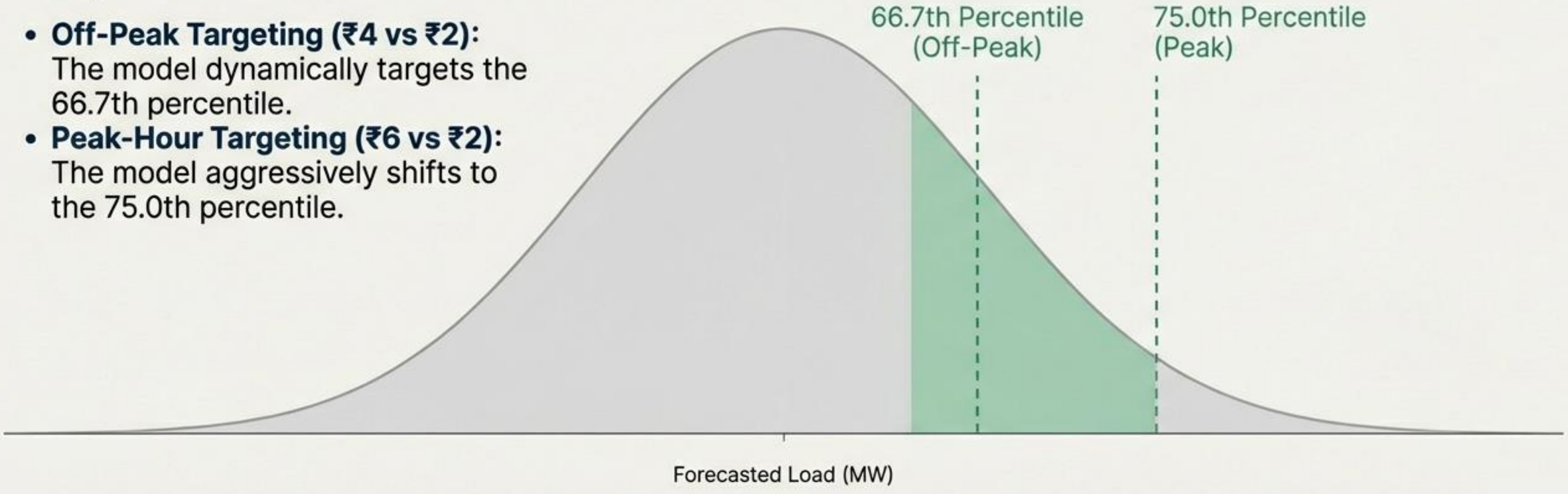


Engineering financial resilience with Newsvendor logic

Applying the Newsvendor inventory model to the power grid allows the system to intentionally over-forecast just enough to dodge crippling ₹6/kWh penalties without triggering excessive ₹2/kWh over-over-procurement costs.

$$\textbf{Optimal Quantile} = \frac{\text{Cost of Under}}{\text{Cost of Under + Cost of Over}}$$

- **Off-Peak Targeting (₹4 vs ₹2):** The model dynamically targets the 66.7th percentile.
- **Peak-Hour Targeting (₹6 vs ₹2):** The model aggressively shifts to the 75.0th percentile.



Precision feature engineering and regime modelling

Strict Issuance-Time Discipline

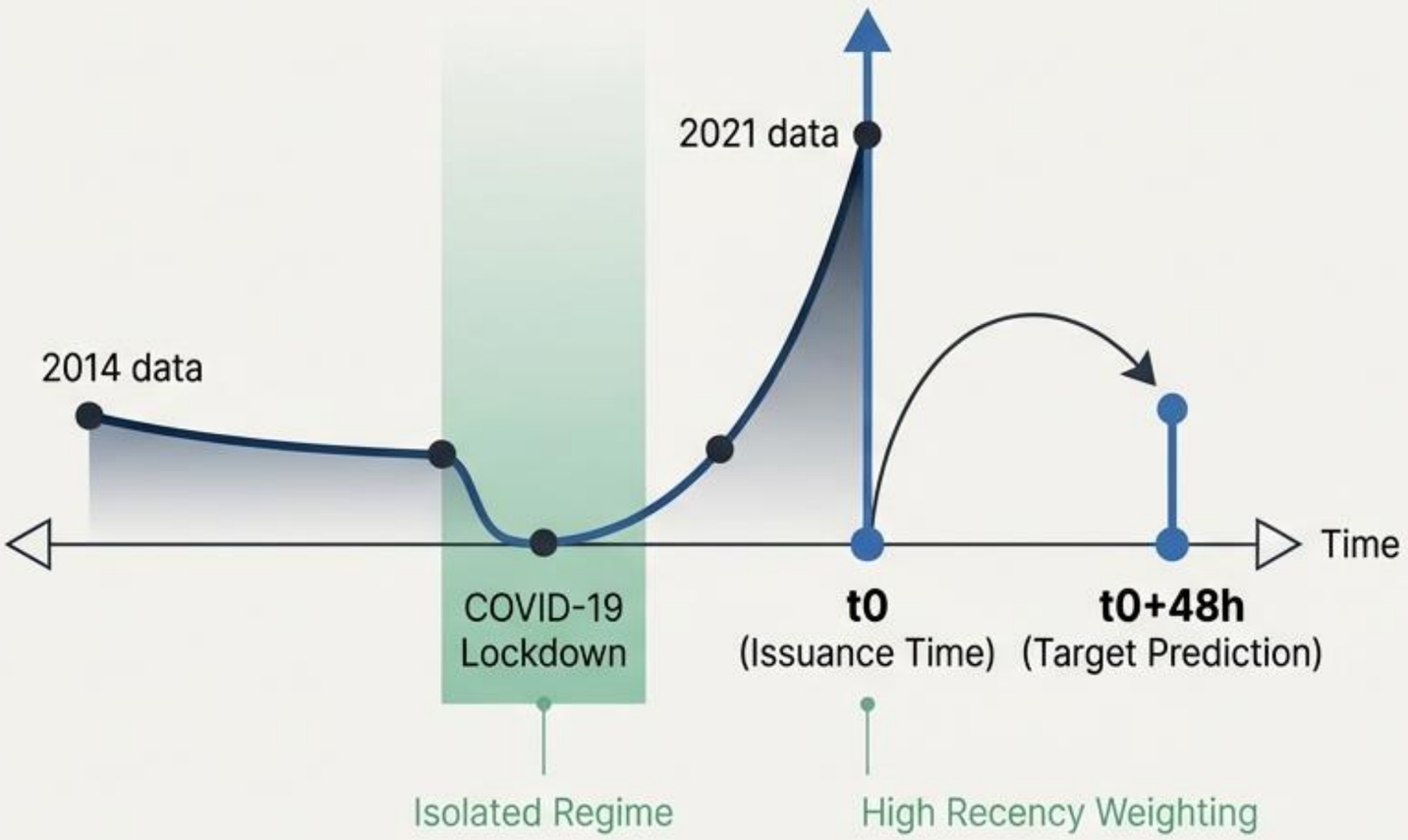
To prevent data leakage, t0+48h predictions rely strictly on t0 safe data (calendar, weather) and 48-hour/1-week physical lags.

Regime Isolation

Explicitly isolated COVID-19 anomalies separating the lockdown and recovery phases.

Log-Scaled Recency Weighting

The model explicitly trusts 2021 hybrid work elasticity over outdated 2014 baseline data, applying exponential decay to historical relevance.



Stage 1 validation confirms a 34% reduction in exposure

Naive Baseline: ₹13,109,175

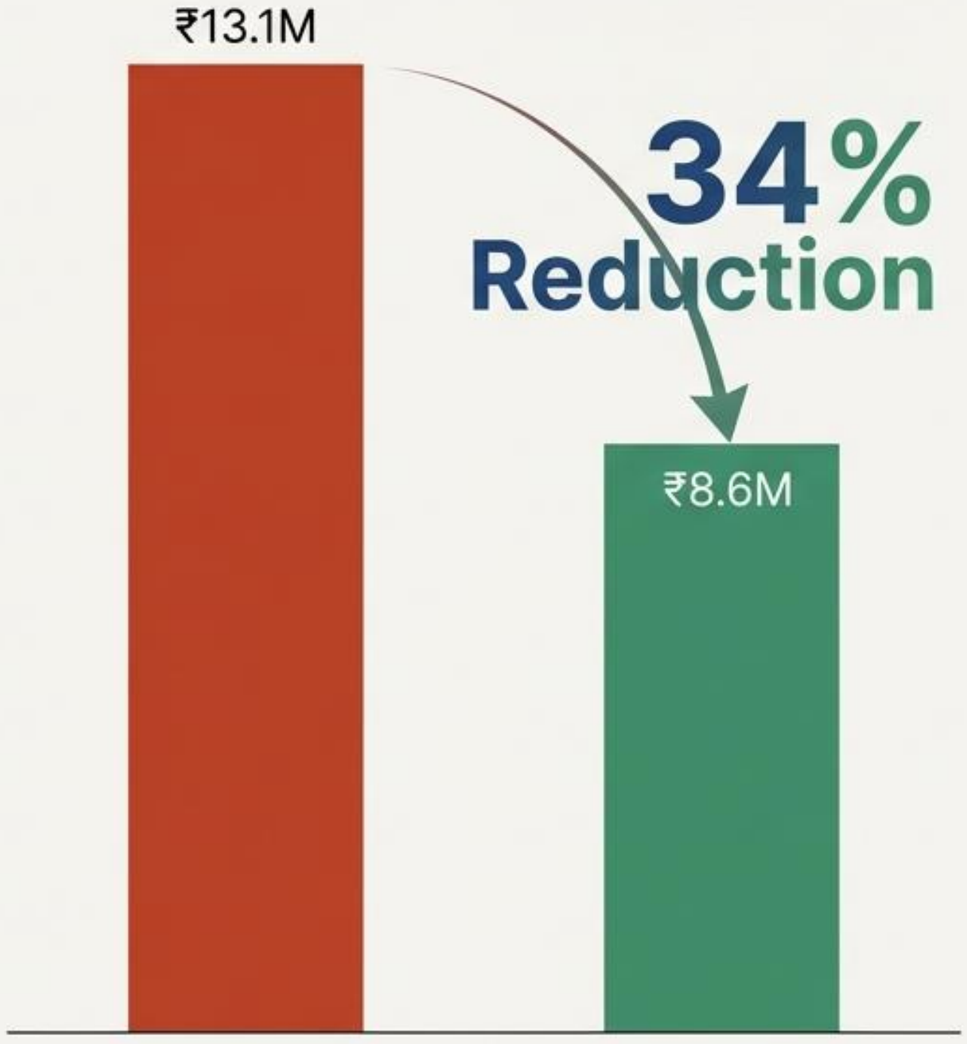
Risk-Optimised Strategy: ₹8,644,653

Value Delivered

34.06% reduction in total financial exposure.

Risk Managed

The 95th Percentile Absolute Deviation was tightly contained to 184.23 kW, proving stability alongside cost savings.



Escaping the ₹6/kWh peak penalty shock

The SLDC Escalation

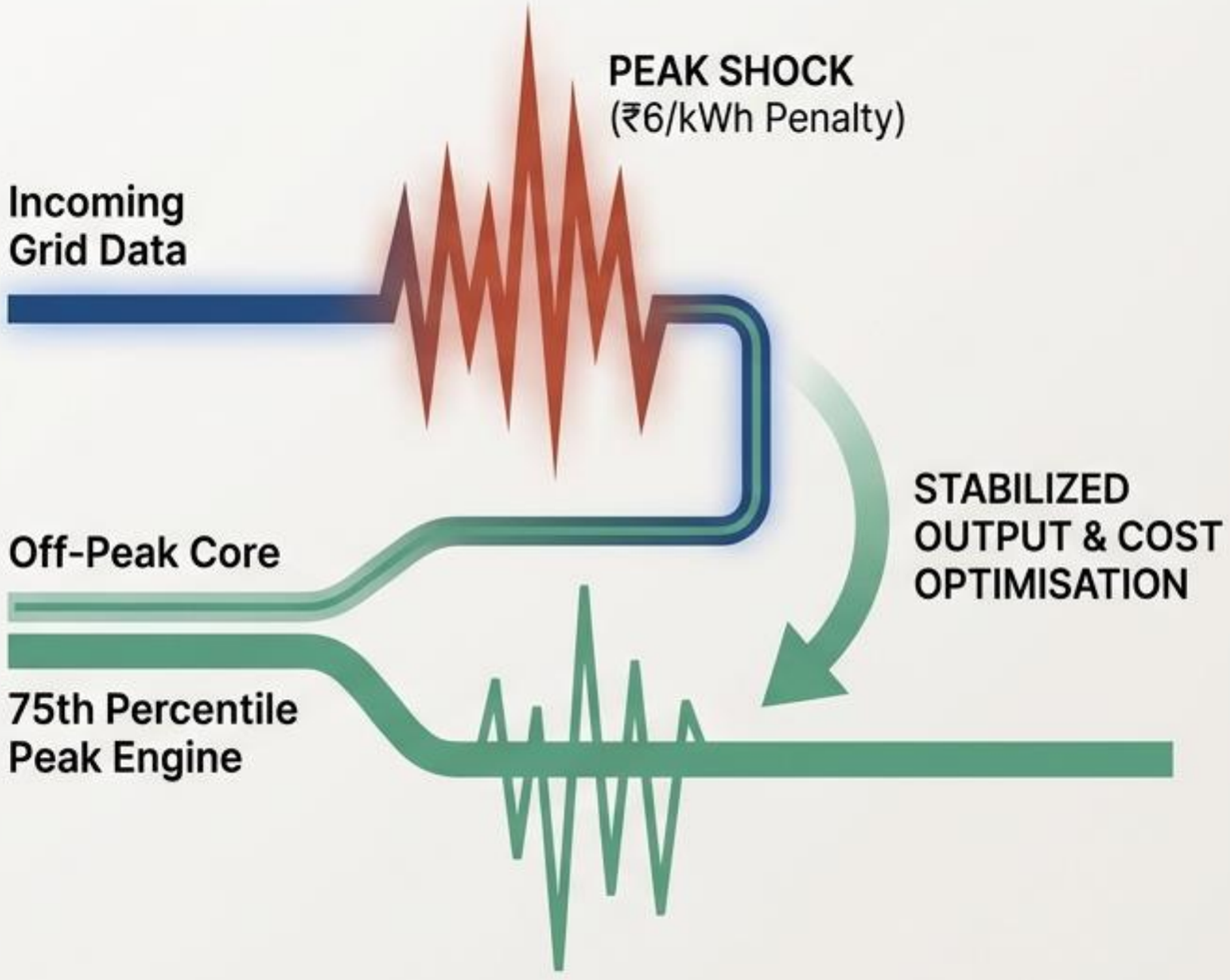
The State Load Despatch Centre immediately raised the peak under-forecasting penalty to ₹6/kWh on unseen, highly volatile test data.

The Danger

Uncalibrated models would suffer catastrophic financial bleeding between 18:00 and 22:00.

The Response

Seamlessly deployed a bifurcated architecture, activating a secondary LightGBM model trained explicitly on the 75th percentile to absorb the intraday shock.

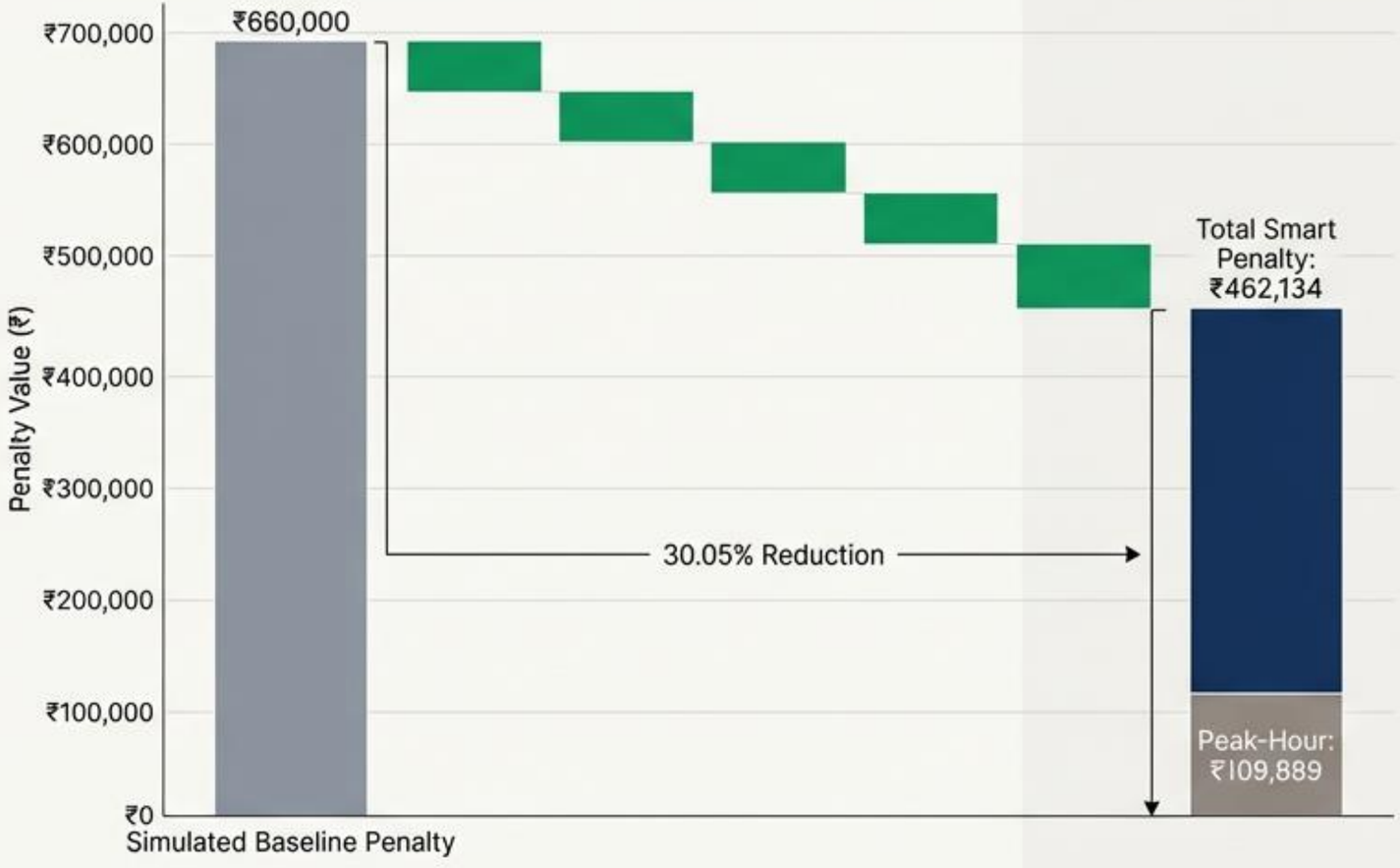


Outperforming the peak-hour shock

Empirical Results on the Out-of-Time Test Set:

Total Smart Penalty: ₹462,134

Peak-Hour Component: Tightened to just ₹109,889, successfully mitigating the ₹6/kWh risk.



Value Delivered

Maintained a 30.05% penalty reduction against the baseline, completely neutralising the heightened regulatory aggression and amplified data volatility.

The confidential board directives

Financial minimisation is necessary, but grid reliability is mandatory.
The architecture must clear four strict hurdles:

1 Total Financial Exposure

Stay strictly below the baseline penalty cap.

2 Peak Reliability

Peak Underestimation > 5% limited to a maximum of 3 intervals.

3 Overall Bias Bounds

Overall Bias mathematically restricted to a [-2%, +3%] window.

4 Forecast Uplift

Average forecast uplift firmly capped at 3%.

The post-processing mathematical equilibrium

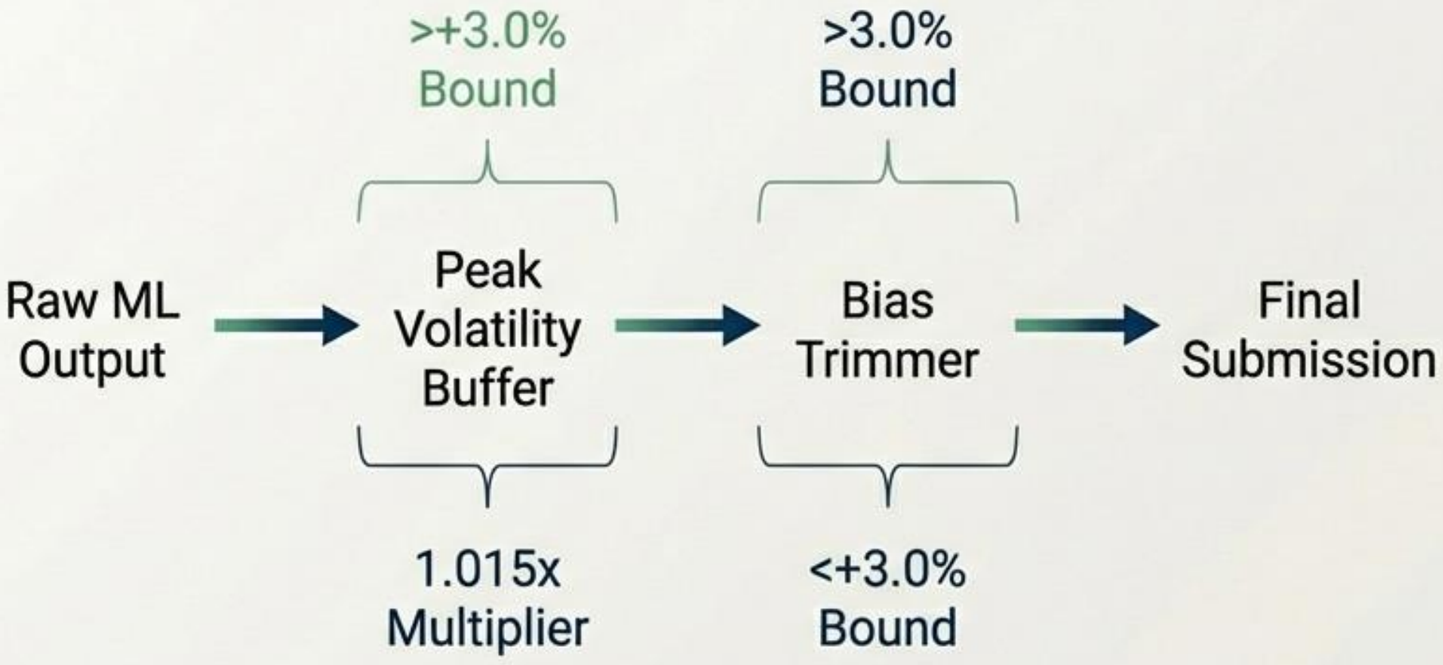
This perfectly satisfies the overall bias constraint without stripping the protective buffering from the peak hours.

Peak-Hour Reliability

Programmed a 1.015x micro-buffer explicitly applied only during the volatile 18:00–21:45 window to prevent tail-risk breaches.

Mathematical Bias Trimming

A pre-submission gatekeeper dynamically scales the final vector to pull overall bias down to exactly <+3.0%.



100% compliance across all board directives

We have transformed a volatile regulatory penalty mechanism into a predictable, optimised, and compliant financial operation.

Total Exposure

Minimised and successfully capped beneath the baseline.

Peak Reliability

<3 interval breaches guaranteed via targeted 1.015x buffering.

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<3 interval breaches guaranteed

Overall Bias

Mathematically locked within the [-2%, +3%] bounds.

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Mathematically locked within the [-2%, +3%] bounds.

Forecast Uplift

Capped firmly within the 3% limit.

Governing the exceptions and risk transparency

No model is flawless; professional risk management requires governing the exceptions.

Root Cause

The highest volume deviations occurred during unpredicted meteorological anomalies (sudden cooling) where momentum lags failed to catch up.

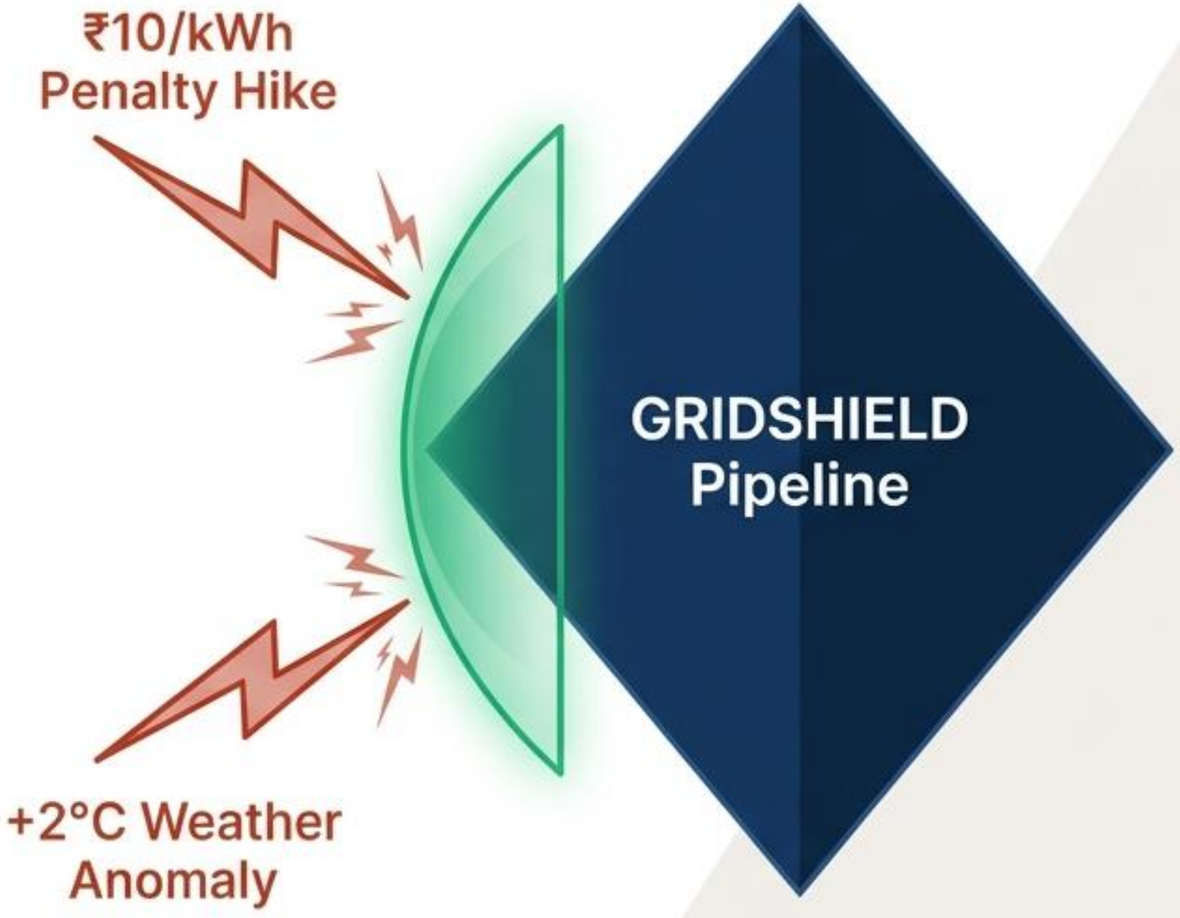
The Financial Armour

Because of our asymmetric positioning, these rare errors manifested primarily as over-forecasts. They incurred only the ₹2/kWh penalty, completely shielding the grid from the severe ₹6/kWh under-forecast hits.

Worst 5 Deviation Intervals

Interval	Forecast (MW)	Actual (MW)	Deviation (MW)	Deviation Direction
19:30-19:45	1250	1195	55	Over-forecast
20:00-20:15	1310	1260	50	Over-forecast
19:45-20:00	1285	1240	45	Over-forecast
20:15-20:30	1340	1298	42	Over-forecast
19:15-19:30	1220	1180	40	Over-forecast

Stress-testing the pipeline against future shocks



Shock 1: Penalties rise to ₹10/kWh

The object-oriented code architecture allows instant dynamic shifting to the 83.3rd percentile without requiring the pipeline to be retrained.

Shock 2: Weather Forecast Errors (+2°C)

By heavily weighting the 48-hour physical momentum lag over pure meteorological regression, the system is insulated from systematic weather forecast failures.

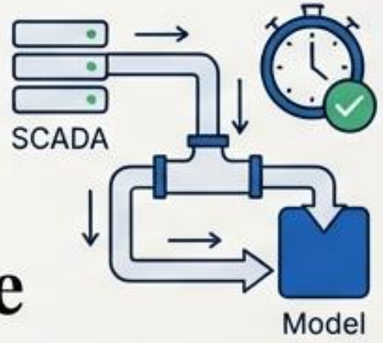
Constraint Tensions

Aggressively pursuing 0 peak errors violates the +3% bias cap. Our micro-buffer remains the proven mathematical equilibrium.

Strategic deployment roadmap

01. Deploy

02. Automate



03. Monitor

1. Deploy the Dual-Quantile Engine

Move the LightGBM models into production, utilising the post-processing optimisation layer to guarantee constraint compliance.

2. Automate Real-Time SCADA Integration

Automate the ingestion of 48-hour lag features to completely eliminate t0 manual data wrangling.

3. Continuous Regime Monitoring

Establish an automated variance alert system to detect new structural shifts (e.g., massive EV charging integration) to trigger dynamic recency-weighting updates.